

Construction and Practice of the Music Curriculum System in Artificial Intelligence Scenes for Normal Majors in Higher Vocational Colleges

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ABSTRACT

With the development of artificial intelligence, traditional music curricula for music education majors at Chinese vocational colleges can no longer meet diverse teaching needs. This article focuses on the deep integration of artificial intelligence and music teaching; through in-depth analysis of existing music teaching modes, it finds shortcomings in traditional curricula. Based on this analysis, the article constructs a music curriculum suitable for the actual needs of Chinese vocational colleges and highlights the advantages of artificial intelligence technology. The actual effect of the proposed curriculum was verified with experiments showing that the newly constructed music curriculum effectively stimulated students' learning enthusiasm, improved students' music skills and practical abilities, and improved students' innovative thinking and overall quality. This study provides a reference for curriculum construction in higher vocational normal colleges and has theoretical significance and practical value.

KEYWORDS

Higher Vocational Colleges, Normal Class, Artificial Intelligence, Music, Curriculum System, Construct

INTRODUCTION

In recent years, new technology based on artificial intelligence (AI) has profoundly changed all aspects of education, and music education is no exception. With the development of digitalization and intelligence, music education is gradually entering a new era (Lian, 2023; Liu, 2025a). In this process, the traditional music education curriculum is facing unprecedented challenges and opportunities. Especially in vocational colleges, there are problems in the traditional curriculum for music majors, such as a single teaching mode, lack of attention to students' subjectivity, and lack of teaching interaction, which seriously restrict the development of students' professionalism and ability to innovate (Fu, 2025; Wang, 2025). At the same time, society's demands for the comprehensive quality and practical teaching ability of graduates are increasing, and the traditional teaching mode struggles to meet current needs (Liu et al., 2024; Okolie et al., 2019).

To discuss the reform of music education in higher vocational normal colleges, it is particularly important to understand the application of AI in the music field (Chen & Sun, 2024). At present, AI

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not only shows extraordinary ability in intelligent composition and arrangement using tools such as Artificial Intelligence Virtual Artist and MuseNet, which can generate creative music works (RaNa, 2025; Wei et al., 2025), but also achieves a breakthrough in personalized learning experience by providing students with customized learning paths and instant feedback through an intelligent music education system that greatly improves learning efficiency and personal satisfaction (Lu, 2025; Peng, 2025a). In addition, AI-based music emotion recognition technology provides new possibilities for deepening music aesthetic education, making the teaching process more accurate, efficient, and individualized (Li & Sun, 2023).

Faced with this background, how to effectively integrate AI technology to build a more interactive and personalized music education curriculum has become one of the important issues to be solved urgently in higher vocational normal colleges (Dai, 2021). Although many scholars have published fruitful research in the field of combining AI with music teaching, most of these studies have focused on art majors at ordinary universities or on music education in primary and secondary schools; such studies have paid relatively little attention to the uniqueness of higher vocational normal universities. The students trained by such colleges have a clear orientation toward teaching practice; students must not only master musical knowledge but also skillfully use new technologies for classroom teaching practice. This type of education has more complex and difficult requirements for the construction of its curriculum (Partti et al., 2021; Yu et al., 2023).

To consider the practical difficulties and future development of music education in higher vocational normal colleges, this article proposes a music curriculum construction scheme based on AI. This curriculum aims at promoting both students' practical ability and their innovation ability through practical exploration and provides new ideas for the modernization of higher vocational education. This study will deeply explore the potential for applying AI technology in music education, analyze new ways to optimize teaching methods for college music, introduce the design concept of an intelligent music education system and its influence on students' learning achievements, and discuss the potential value and development prospects of digital intelligence courses for improving teaching quality and cultivating musical abilities. In addition, a music performance system based on an emotion recognition algorithm will be studied to lay a foundation for the development of AI-based emotion perception modules in music education. Through the above efforts, I hope to contribute to the innovation and reform of music education in higher vocational normal colleges and to fill gaps in related research fields.

The remainder of this article is arranged as follows. The next section comprehensively summarizes relevant literature on the integration of AI into music education, highlighting existing achievements and research gaps. The next part expounds on the theoretical basis and research methods of the article. Then I discuss the verification design, experimental process, and teaching effects of a proposed AI-based music education curriculum. Next, I discuss the research findings, acknowledge the limitations of these findings, and put forward conclusions and suggestions for future development. Finally, the conclusion summarizes the research results and contributions of this article.

LITERATURE REVIEW

In recent years, research on the integration of AI into education has gradually become the focus of Chinese and international educators. At present, the results of research into AI music education have been rich, and various teaching models have emerged to promote teaching efficiency. Johnson (2020) solved the limitations of moving teaching from a face-to-face environment to an online environment by defining an innovative conceptual model. The key factors to be considered included teaching methods, learning methods, the online technology environment, and students' skills and knowledge. Wu (2023) took vocal music teaching as the main object, introduced several types of intelligent algorithms to construct an evaluation workflow, and proposed a comprehensive intelligent evaluation model for vocal music teaching based on mixed intelligence. The model adopted a convolutional neural network, long

short-term memory network, and multilayer perceptron to establish a novel integrated structure as the main evaluation model. However, for practical applications at higher vocational normal colleges, these advanced models still have obvious shortcomings: weak guidance for teaching practice, narrow application scope, high technical barriers to entry, and great difficulties in implementation.

Peng (2025b) pointed out that the application of deep learning (DL) algorithms in music education is helpful to meet the learning needs of students of different ages and musical literacy levels. This study introduced the basic principles and application status of AI and DL, integrated them into the design of music education, constructed a corresponding algorithm model, and wrote the necessary code. The experimental results showed that the music emotion recognition system based on DL had higher recognition accuracy, improved the effect of music education, and provided new possibilities for future teaching activities. This achievement emphasized the potential of AI technology in improving the quality of music education, especially in emotion recognition and personalized feedback.

Lu (2025) discussed the integration of a machine learning system into an intelligent music education system for real-time feedback and performance evaluation. This system integrates DL technology that enables it to analyze elements such as pitch, rhythm, strength and expressive force, then to give students suggestions for improvement. The research results proved that equipment based on machine learning can change the rules of traditional music teaching and creativity and can provide a broader and more friendly framework for modern music education.

Liu (2025b) discussed the digital and intelligent development of AI-based higher music courses from the perspective of curriculum construction. The article pointed out that an intelligent music course should pay attention not just to the cultivation of musical theory and practical ability but also to integrating a personalized and interactive teaching system. Although AI has shown great advantages for intelligent composition, real-time feedback, and virtual performance, the shortage of teachers and the uneven distribution of resources are the main obstacles preventing its deep integration. This study put forward core construction elements and innovative strategies and emphasized the strategic significance of the course of digital intelligence in improving teaching quality and cultivating musical ability.

The research of Wang and Guo (2021) focused on emotional differences of listeners in complex music performance systems and proposed an emotion recognition and classification algorithm based on the fusion of multiple classifiers. By comparing support vector machine, k -nearest neighbor, artificial neural network, and Iterative Dichotomiser 3 classifiers, they found that the fusion of a support vector machine and an artificial neural network was the best for audio signal classification after preprocessing; the F_1 value and recall rate were better than for a single model. Their study provided an efficient algorithm combination scheme for the recognition of music emotion and laid a foundation for the development of an AI-based emotion perception module in music teaching.

Tian (2025) systematically discussed how intelligent technology can support the development of music professionals, focusing on its role in the analysis of musical structure, reconstruction of creative ability, and transformation of the teaching system. The research revealed how technology can innovate musical idioms and creative models.

Ng et al. (2022) discussed using the musical instrument *myu* in the online flipped classroom by learning *Shubailan* (a form of folk singing). Their research allowed students to participate in course preparation through preset recorded videos and collaborative activities in online classes. They found that the online flipped classroom method improved students' satisfaction and knowledge acquisition related to the process of Shubailan music creation and provided suggestions for online music educators.

RaNa (2025) reviewed the application of existing AI tools in music creation to enhance on-site activities and improve how people and machines jointly create music. By analyzing the case studies of MuseNet, Magenta, Artificial Intelligence Virtual Artist, and musicians using AI, the role of AI in creativity within the music industry was studied and discussed. This research stated that AI will assist musicians in the future, not replace them.

However, most of these studies have focused on the application of AI technology in ordinary universities or art colleges and have not considered the unique teaching situation of higher

vocational normal universities. Despite these rich achievements, research on the deep integration of AI technology into higher vocational normal music education has not been fully carried out and needs further exploration and innovation. Therefore, this article attempts to integrate the most recent theories of AI with the actual needs of higher vocational normal majors. In this article, I put forward a new path for curriculum construction, with an objective of filling existing research gaps and providing effective reference to higher vocational normal music education.

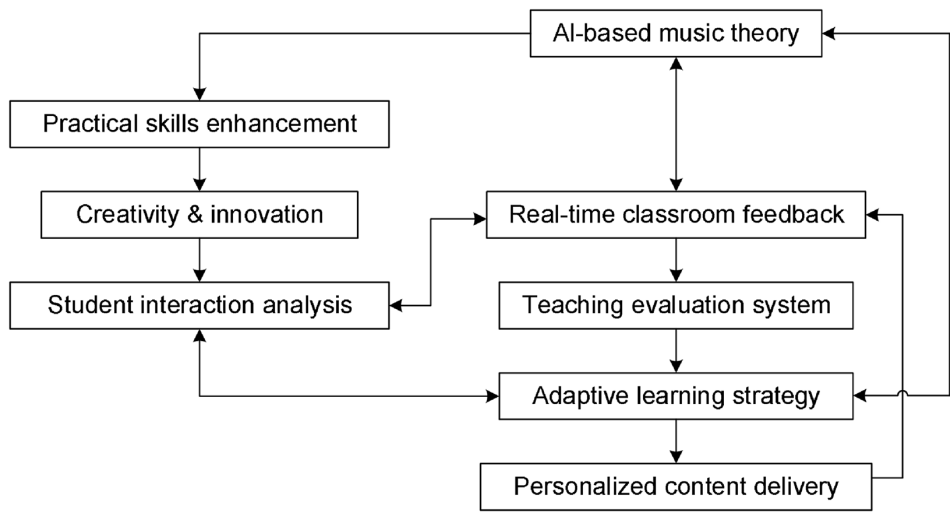
MATERIALS AND METHODS

Theoretical Basis of AI and Music Education

AI technology has rapidly penetrated into the field of education in recent years, and many studies have pointed out that it is outstanding at improving learning interest and efficiency. Specific to the field of music education, the intervention of this technology has also brought about a revolutionary change (Li & Wang, 2024). On the one hand, by relying on big data analysis and machine learning, AI technology can effectively identify and evaluate students' musical ability and learning style, thus realizing a personalized education process (Hou, 2024; Tabak, 2023). On the other hand, the intelligent interactive system can also enhance the interaction between teachers and students, stimulate students' creative thinking, and make up for the shortcomings of traditional teaching interactions (Ng et al., 2022). The formation of this theoretical basis has obviously broadened the depth and breadth of music education and promoted the transformation from a traditional mode to an intelligent mode.

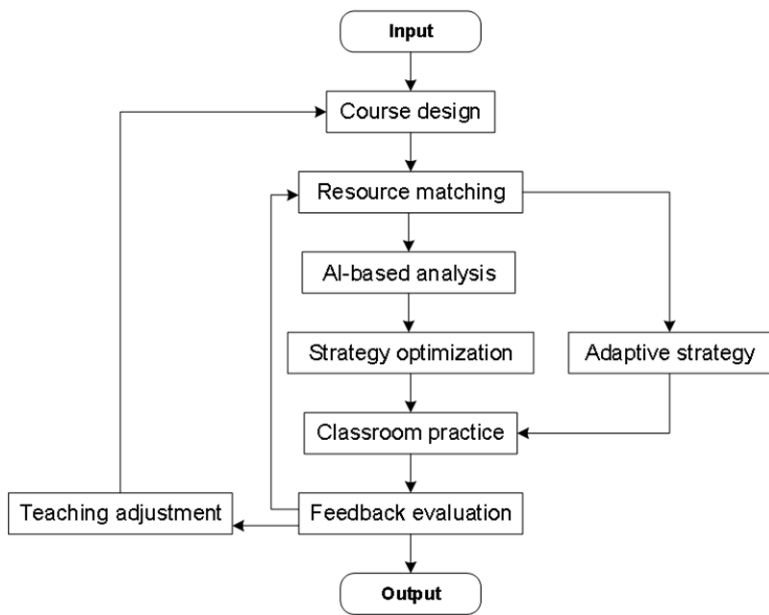
Although the theoretical prospect of this mode is very bright, the use of AI-based tools is not smooth in practice. At the present stage, AI-based music education for education majors in higher vocational colleges still faces many practical difficulties. For example, the existing teaching model relies too much on the preset programmed mode and ignores the dynamic needs of normal students' teaching practice (Guan, 2023). In addition, the current model design pays more attention to the complexity of the technology itself and often ignores the operability of the actual teaching application environment, resulting in poor effectiveness (Hu, 2025). Therefore, difficult problems in music education cannot be completely solved just by relying on technological progress, and there is still a gap between theory and practice that cannot be ignored. Addressing this gap requires a more interactive and feedback-based theoretical model to combine AI with music education to meet the special needs of higher vocational normal students. Figure 1 depicts a theoretical framework for the integration of AI into music education, emphasizing a circular feedback mechanism between theory and practical teaching to improve the adaptability and practicality of the model through two-way interaction.

Figure 1. Theoretical Framework for Integration of AI Into Music Education



With AI-based music theory as the core, this model connects real-time classroom feedback through two-way interaction, drives a teaching evaluation system and adaptive learning strategy, and realizes personalized content delivery. At the same time, the system feeds back to creativity, innovation, and student interaction analysis through practical skills enhancement, forming a closed-loop structure that continuously optimizes teaching strategies and improves students’ comprehensive ability.

Figure 2. Overall Structure and Feedback Process of the Proposed Model for a Music Education Curriculum



In this model, the feedback and adjustment links form a complete, closed-loop structure that greatly improves the adaptability and operability of teaching. The process framework starts with input and enters the core loop through course design, which drives strategy optimization through resource matching and AI-based analysis and jointly guides classroom practice with adaptive strategy. After practice, evaluation results are generated by feedback evaluation, which forms the output and triggers teaching adjustment to feed back into the course design. In addition, the results can be fed back to resource matching for adjustment. This process forms a dynamic closed-loop optimization mechanism to realize continuous iteration and personalized adaptation of teaching strategies.

Proposed Construction of AI-Based Music Curriculum in Higher Vocational Colleges

To effectively solve the existing problems of music education in higher vocational normal colleges, this article designs a music curriculum system integrating AI technology. This system is different from the traditional prescriptive teaching mode because it highlights the initiative and personalized needs of students. Specifically, through the intelligent collection and analysis of students' learning data, the system realizes accurate matching and pushing of teaching resources to meet the individual differences of students. The specific personalized matching degree is shown in this formula,

$$M = f(S, T, P) \quad (1)$$

where M represents the degree of personalized matching, S represents the parameters of students' personality characteristics (such as study habits and interests), T represents the parameters of teaching content, and P represents the parameters of teaching strategies. The design of the formula aims to clearly and objectively reflect the degree of adaptation between the course content and the students' needs to help teachers implement the teaching plan in a targeted manner and to significantly enhance the students' enthusiasm toward learning.

At the same time, this study also emphasizes the function of real-time feedback and adjustment in the teaching process. This mechanism is based on real-time performance in the classroom and adjusts teaching methods to achieve the best teaching effect. The specific adjustment mechanism can be expressed as this formula,

$$R_{\text{new}} = R_{\text{old}} + \alpha(E_{\text{desired}} - E_{\text{actual}}) \quad (2)$$

where R_{new} is the updated teaching strategy parameter, R_{old} represents the original teaching strategy parameter, E_{desired} is the expected teaching effect, E_{actual} is the actual teaching effect, and α is the adjustment coefficient. This real-time and dynamic adjustment mechanism ensures that teaching strategies are more suitable to students' actual needs and makes the teaching process more flexible and efficient.

To achieve accurate personalized matching, the system first established a multimodal data acquisition layer covering auditory, visual, and behavioral interactions. In the aspect of audio signal acquisition, the system captured data on the students' pitch, rhythmic stability, timbre, dynamics, and sustain in real time through a high-fidelity microphone with a sampling rate of 48 kHz. In addition, vision and motion capture technology were introduced; a camera and the OpenPose computer vision algorithm were used to analyze students' sitting posture, hand structure, and body rhythm in real time, thus addressing the problem that it is difficult to quantify performance posture in traditional teaching. Finally, the system recorded an interactive behavior log during the whole process, including practice duration, track repetition times, error correction response time, and other metrics that provided a basis for subsequent analysis.

For the core data processing layer, a convolutional cyclic neural network hybrid architecture was adopted. This convolutional neural network, a variant of ResNet-18, was responsible for extracting local features from the audio spectrogram, while a bidirectional long short-term memory network was used to process time series data to analyze the coherence and fluctuation of phrases. The key parameters of this module were set as follows: The learning rate was 0.001 (using Adam optimizer), the batch size was set to 32, and a dropout ratio of 0.3 was introduced to prevent overfitting. This architecture can capture both instantaneous pitch deviation and long-term musical expression.

Figure 2 depicts the overall structure and feedback process of the proposed model.

Verification of the Curriculum in Practice

To scientifically evaluate the actual effects of the curriculum proposed in this article, this study carried out strict practical verification in the real classroom environment of higher vocational normal colleges. When verifying the design of the scheme, three important teaching indicators—students' practical musical skills, creativity, and classroom interaction—were considered to make a comprehensive and objective analysis of the effectiveness of the new system. In the specific implementation process, I selected 90 students from three classes as research participants, strictly implemented the new system in the teaching practice for one semester, and regularly recorded and evaluated relevant data. For quantitative analysis, this study used three quantitative indicators—classroom performance score, professional skills assessment, and student self-assessment scale—to obtain reliable evaluation data.

RESULTS

Analysis of the Current Situation of Music Education in Higher Vocational Teachers' Colleges

In recent years, normal music education in higher vocational colleges has made great progress, but with the iterative upgrading of educational technology, many problems have also been exposed. First, the current teaching mode is relatively simple, and teachers mainly adopt the traditional teaching mode: Teachers talk and students listen. Although this mode can improve students' theoretical knowledge in the short term, it is difficult to cultivate students' teaching practice skills and innovative thinking, which may even dampen students' enthusiasm for learning in the long run. In addition, there is a certain lag in curriculum design at this stage, which cannot keep up with rapid changes in educational technology and practical application in time, resulting in a disconnect between theory and practical instruction.

In addition, through the actual investigation of normal majors in several higher vocational colleges, it is found that the digital technology used in teaching in most colleges is extremely limited, and intelligent equipment is even more scarce. Table 1 reveals the usage ratio, teacher satisfaction, student satisfaction, and difficulty in promotion of different teaching interaction modes. The data in Table 1 are obtained through field investigation of conventional majors in several higher vocational colleges. This investigation focuses on the application of digital technology and the allocation of intelligent equipment in the teaching process of colleges.

Table 1. Survey Data on Music Teaching Interaction Modes in Higher Vocational Normal Schools

Teaching interaction mode	Usage ratio (%)	Teacher satisfaction (%)	Student satisfaction (%)	Difficulty in promotion (score of 1–10)
Traditional teaching	75.2	68.5	65.4	2.0
Group discussion	45.6	75.3	78.2	4.0
Digital interaction	28.5	72.6	80.1	7.0
Online interactive platforms	15.4	70.4	82.5	8.0
AI-assisted teaching	8.7	62.1	85.7	9.0

The data show that the usage ratio of traditional teaching was the highest, reaching 75.2%, but that teacher and student satisfaction were relatively low (68.5% and 65.4%), and the difficulty in promotion was the least (2.0 points). In contrast, AI-assisted teaching had the highest student satisfaction (85.7%), but the usage ratio was only 8.7%, and it faced the greatest resistance to promotion (difficulty in promotion is 9.0 points), mainly because of technical limitations. Group discussion, digital interaction, and online interactive platforms showed an increasing distribution in usage ratio and satisfaction. Among these measures, with a higher degree of digitalization, students’ satisfaction increased, but the difficulty in promotion also increased.

The above results demonstrate that the application ratio of AI technology was very small, which led to a lack of interaction and innovation in teaching. The current music education curriculum in normal colleges and higher vocational colleges is in urgent need of deep-seated reform to meet the needs of current teaching practices. More importantly, music education should not just stay on the surface of theoretical instruction. What really needs to be done is to encourage students to participate in teaching practice through multiple channels inside and outside the classroom, in order to cultivate music teachers in the new era with innovative ideas and practical abilities.

Analysis of Existing AI Music Education Models

In the past, many AI-based music education models paid more attention to the innovation and complexity of the technology itself, ignoring the actual needs and operating environments of users, which led to the frustration of teachers and students. This study evaluated existing AI music education models and gathered performance comparison data summarized in Table 2.

Table 2. Comparison of Performance of Existing AI-Based Music Education Models

Type of model	Feedback efficiency (%)	Practice relevance (score of 1–100)	Technical ease of use (score of 1–100)	User satisfaction (score of 1–100)	Cost input (score of 1-10)
MusicGenie intelligent interactive creation model	76.0	68.0	55.0	70.2	8.0
MusicSense music emotion recognition model	82.0	71.0	60.0	73.4	7.0
AI-MusicTutor intelligent music evaluation model	78.0	66.0	58.0	68.7	6.0

The results in Table 2 show that the MusicSense music emotion recognition model was the most balanced and led in all performance indicators, with feedback efficiency of 82.0%, practice relevance of 71.0 points, technical ease of use of 60.0 points, user satisfaction of 73.4 points, and relatively low-cost input (7.0 points). The cost input of the MusicGenie intelligent interactive creation model was the highest (8.0 points), while it was slightly inferior in feedback efficiency (76.0%), practice relevance (68.0 points), technical ease of use (55.0 points), and user satisfaction (70.2 points). Although the AI-MusicTutor intelligent music evaluation model had the lowest cost input (6.0 points), its practice relevance (66.0 points) and user satisfaction (68.7 points) ranked last among the three, indicating that low cost did not directly translate into the best teaching effect or user experience.

To sum up, there are some shortcomings in the existing models, which restrict the wide application and promotion of AI in the field of music teaching to some extent. In the future development of AI-based modes of music education, educators must fully consider actual application environments and user needs, lower technical barriers to entry, optimize the interactive process, and truly realize an effective connection between theoretical innovation and practical application.

Verification of the Curriculum in Practice

Through a comparison of data before and after the students’ ability improvement shown in Table 3, the outstanding advantages of the new curriculum in improving students’ practical skills were fully confirmed.

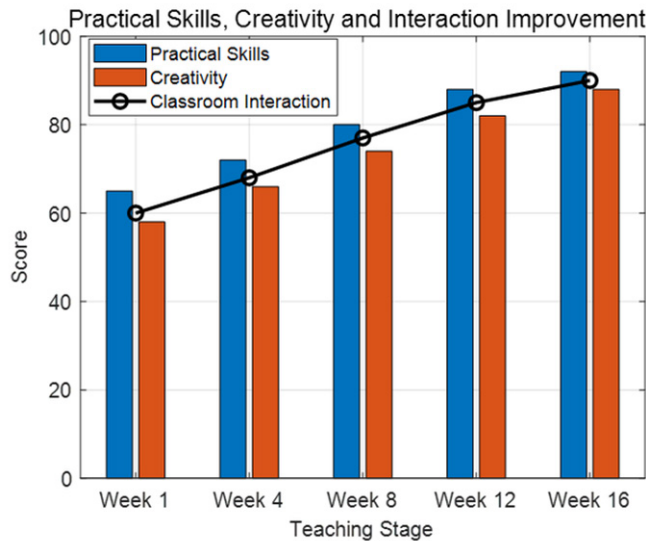
Table 3. Students’ Abilities Before and After Implementation of the Curriculum

Practical ability indicator	Score before implementation	Score after implementation
Musical instrument operation	67	85
Improvisational theater	65	87
Creativity	63	90

The results in Table 3 showed that all indicators improved significantly. In musical instrument operation, the average score jumped from 67 points to 85 points. The ability in improvisational theater greatly improved from 65 points to 87 points. The improvement in creativity was the most significant, with the score soaring from 63 points to 90 points. These data strongly suggest that the proposed curriculum had excellent results in strengthening students’ practical skills, stimulating artistic expression, and cultivating innovative thinking.

Figure 3 shows the dynamic improvement trend of students’ practical skills, creativity, and classroom interaction in the teaching stage from the first week to the 16th week during the implementation of the curriculum.

Figure 3. Improvement Trend of Students' Practical Skills, Creativity, and Classroom Interaction



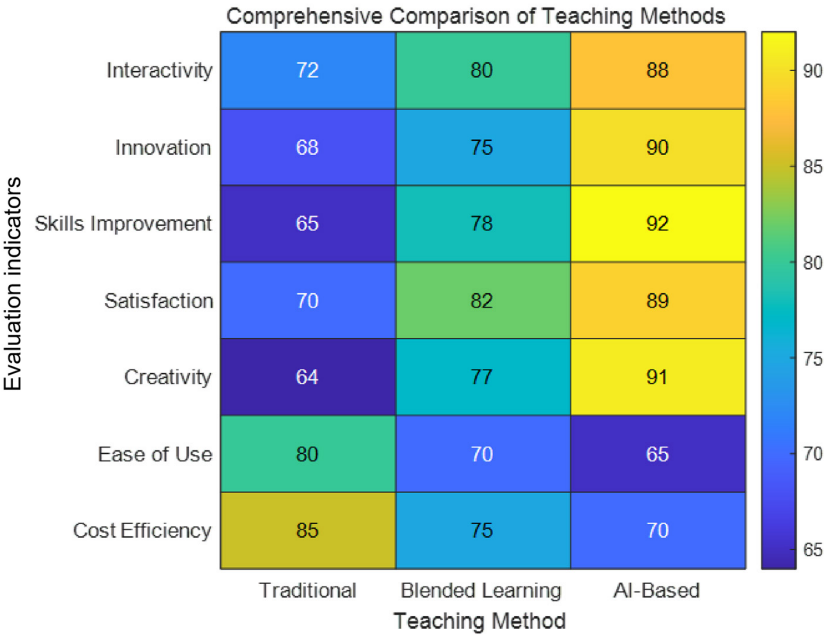
The data show that practical skills increased steadily from 65 points to 92 points and that creativity increased from 58 points to 88 points, both of which showed a continuous upward trend consistent with the expected teaching effect. The classroom interaction score climbed smoothly from 60 points to 90 points, indicating that with the progress of the course, students' participation and collaboration were significantly enhanced. Overall, the three indexes were optimized synchronously in the teaching cycle, which reflects the curriculum's synergistic promotion of comprehensive literacy.

In addition, to ensure the comprehensiveness and objectivity of the verification process, the research also organized several teachers' symposia and student interviews and collected firsthand qualitative materials to understand the subjective feelings and experiences of teachers and students in detail. Students and teachers generally said that the new system significantly enhanced the interactivity and interest of the classroom and made the learning atmosphere more active. However, some teachers mentioned that there are technical challenges in the implementation and recommended that teachers need extra training to better control the application of AI technology. Thus, although practice has proved the effectiveness of the curriculum proposed in this article, it is still necessary to continue to improve teacher training and technical support in the future to ensure the smooth development of teaching practice.

Analysis of Practical Effects and Comparison of Methods

Based on the data obtained in the practical verification stage, this study further analyzed the implementation effects of the AI-based music education curriculum. This study compared traditional, blended learning, and AI-based teaching methods according to several evaluation indicators including interactivity, innovation, skills improvement, satisfaction, creativity, ease of use, and cost efficiency. Figure 4 depicts the performance of these three teaching methods in these seven evaluation indexes in the form of a heat map. Warmer colors (yellow and orange) indicate higher scores, while colder colors (blue and purple) indicate lower scores.

Figure 4. Heat Map of Comprehensive Comparison of Teaching Methods



As can be seen in Figure 4, AI-based teaching methods had an absolute advantage in most measures of educational effect. They received the highest scores on key indicators such as skills improvement (92 points), creativity (91 points), innovation (90 points), satisfaction (89 points), and interactivity (88 points), which were significantly better than with the other two methods. Blended learning was balanced and steady in all dimensions, with scores generally ranging from 70 points to 82 points. The traditional teaching method scored the highest in cost efficiency (85 points) and ease of use (80 points), but it was the weakest in core indicators related to the cultivation of students' ability: skills improvement (65 points), creativity (64 points), and innovation (68 points). This result shows that although AI-based teaching faces certain barriers related to cost and technical requirements, it has unparalleled advantages in improving teaching quality and students' core literacy.

In addition, according to feedback from teachers and students, most students spoke highly of the new system, believing that it effectively improved enthusiasm for learning and participation in class. Teachers generally reported that the new system has obvious advantages with helping them grasp teaching progress and students' situations more accurately. However, it also needs to be admitted that the implementation of the new method has created higher requirements for teachers' ability with information technology, which has caused certain pressure and challenges to teachers. In the future, when further popularizing and applying the new system, more attention should be paid to teacher training and technical support to ensure that all teachers can master the relevant technologies skillfully and finally realize the successful application of the new system.

DISCUSSION

Considering these practical results, the AI-based music curriculum for higher vocational normal colleges proposed in this paper shows strong innovation and feasibility. The results show that the curriculum has obviously improved students' skill performance and innovation ability in practical teaching and has

successfully made up for the shortcomings of lack of interaction and personalized teaching under the traditional teaching mode. This result is particularly worthy of recognition. Traditional teaching has long been criticized because a single teaching model cannot stimulate all students' interest and potential. In addition, the new curriculum is particularly outstanding in dynamically adjusting teaching strategies, which gives it obvious advantages in dealing with diversified teaching needs.

However, there are still some shortcomings in this study, such as the problems of teacher training and technology popularization, which limit the large-scale promotion and application of the new model to some extent. To solve the problem of teachers' lack of information technology ability, this study suggests implementing hierarchical training based on the characteristics of music disciplines, from basic operations to teaching methods based on human-machine collaboration, to help teachers to progress from knowing how to use the technology to making good use of it. In addition, this study suggests establishing a positive incentive mechanism by incorporating AI application ability into teachers' performance appraisals and job title evaluations and by setting up innovation awards and fault-tolerant zones to stimulate teachers' motivation. At the same time, compulsory modules and training sessions in colleges and universities could be established to improve teachers' quality and to realize technical upgrading and teacher development.

In addition, the parameter settings of real-time feedback and adjustment in the curriculum still need to be further optimized in further studies to ensure more operability and accuracy in actual teaching. Although the new curriculum shows great potential, to achieve wide application, it is necessary to continue to improve the technical support system, strengthen teachers' information training, and carry out practical verification in a wider range to further ensure its teaching efficiency and operability.

In addition, the actual implementation of AI-based modes of music education still faces challenges related to data privacy, ethics, and economic cost. In terms of data security, the high-definition audio and real-time video attitude and behavior logs collected by the system are highly sensitive biometric data with a risk of leakage. Therefore, desensitization and feature extraction of the original data must be completed in the local equipment of students, only encrypted and unstructured data should be uploaded, and data access classification should be established in strict accordance with laws and regulations. At the level of economic cost, high-precision intelligent musical instruments, visual analysis terminals, and continuous computing power maintenance costs are often beyond the limited budgets of higher vocational colleges. In this regard, this study suggests adopting cloud services and a bring your own device policy, using students' existing equipment to lower hardware needs, and jointly building and sharing cloud resources through regional colleges and universities to ensure that technology benefits vocational education.

CONCLUSION

This article describes a new AI-based curriculum that addresses the practical problems of courses for music education majors in higher vocational colleges. In the study, the shortcomings of the existing teaching mode are analyzed in detail, and the personalized matching teaching strategy and real-time feedback adjustment mechanism are proposed. The effectiveness and innovation of the system in actual teaching practice are confirmed. The data show that after the implementation of the curriculum system, students significantly improved their practical skills, creativity, and classroom interaction, and that the teaching effect is better than that of the traditional model. In addition, the teaching model and feedback mechanism designed in this paper reflect good adaptability and flexibility; they can effectively cope with the complexity of the teaching environment of higher vocational teachers.

From the perspective of research value, the curriculum proposed in this paper enriches the theoretical basis of music education and provides a practical teaching method model with strong practical significance. Nevertheless, in its specific application, the system puts forward higher requirements for teachers' information technology ability, and there can be barriers to the system's adoption. Future research needs to focus on solving this bottleneck, such as increasing information

technology training for teachers, reducing the difficulty of using the system, and further improving the usability of the model. At the same time, the accuracy and real-time performance of teaching feedback mechanisms need to be further improved. In short, the research results provide a new path for the construction of music courses in higher vocational normal colleges and provide a broad research space for future educational practice.

COMPETING INTERESTS

The author of this publication declares there are no competing interests.

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